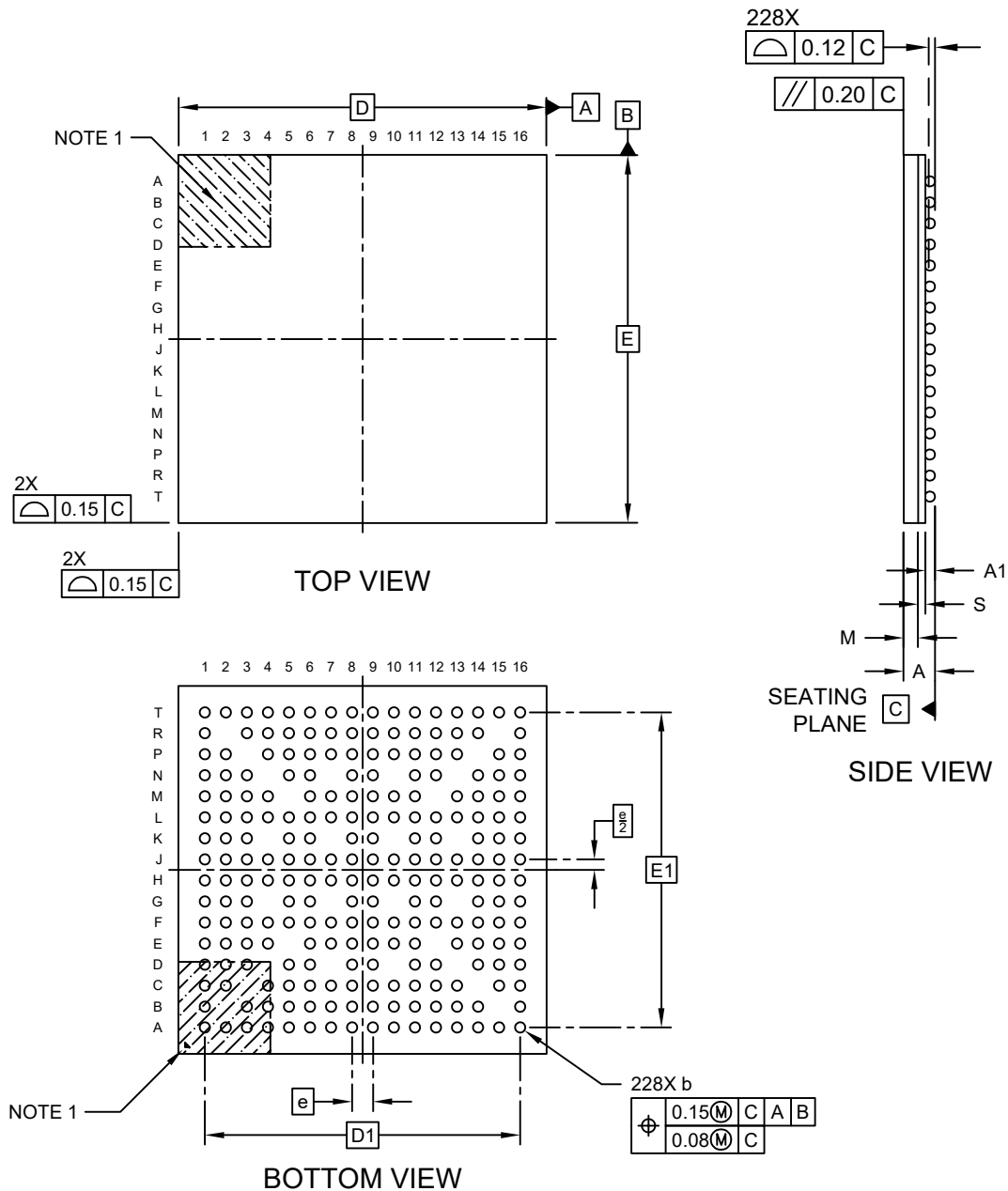


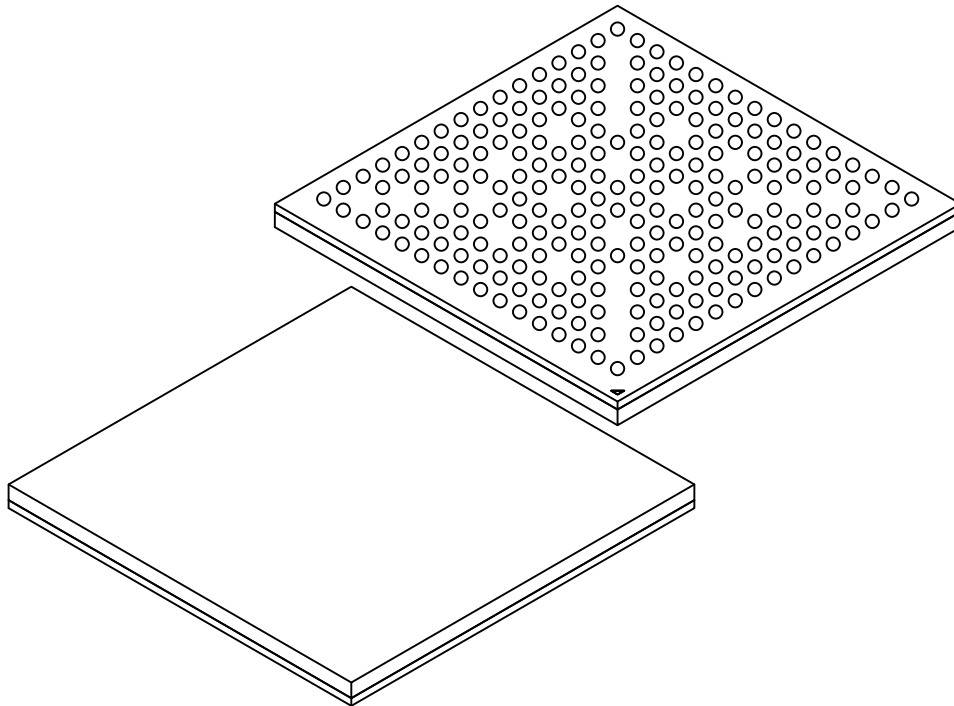
228-Ball Thin Fine Pitch Ball Grid Array (HGB) - 14x14x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



228-Ball Thin Fine Pitch Ball Grid Array (HGB) - 14x14x1.2 mm Body [TFBGA]

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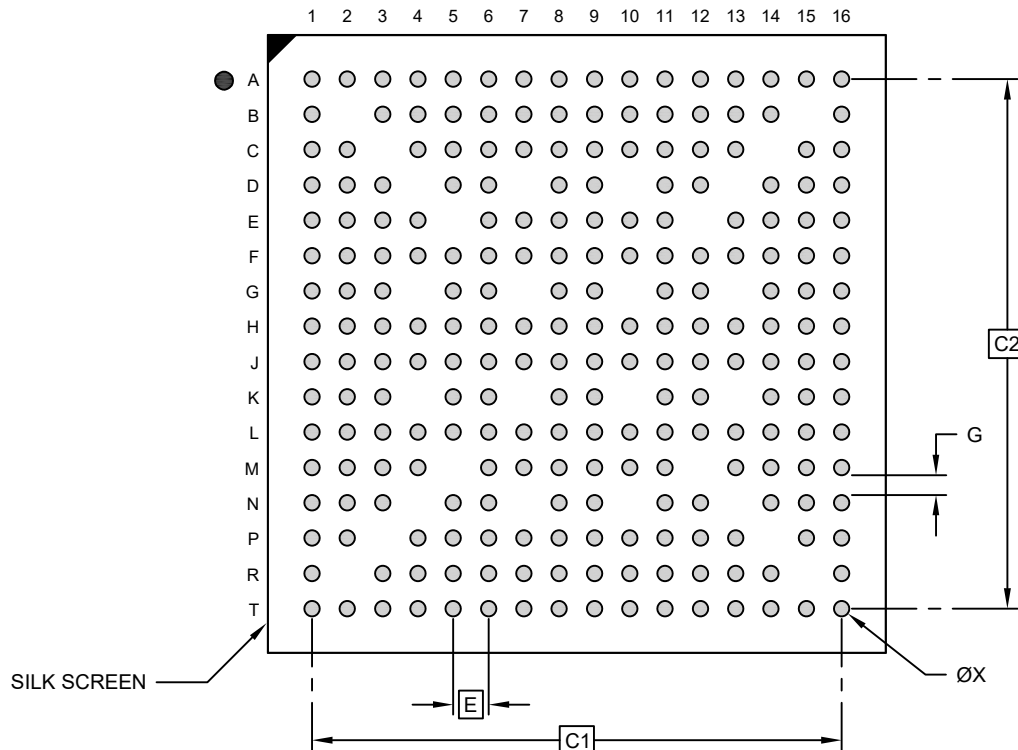
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	228		
Pitch	e	0.80 BSC		
Overall Height	A	–	–	1.20
Ball Height	A1	0.27	0.32	0.37
Mold Thickness	M	0.53 REF		
Substrate Thickness	S	0.26 REF		
Overall Length	D	14.00 BSC		
Ball Array Length	D1	12.00 BSC		
Overall Width	E	14.00 BSC		
Ball Array Width	E1	12.00 BSC		
Ball Width	b	0.38	0.40	0.48

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Dimensioning and tolerancing per ASME Y14.5M
BSC: Basic Dimension. Theoretically exact value shown without tolerances.
REF: Reference Dimension, usually without tolerance, for information purposes only.

228-Ball Thin Fine Pitch Ball Grid Array (HGB) - 14x14x1.2 mm Body [TFBGA]

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



RECOMMENDED LAND PATTERN

Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Contact Pitch	E	0.80 BSC		
Contact Pad Spacing	C1	12.00 BSC		
Contact Pad Spacing	C2	12.00 BSC		
Contact Pad Diameter (X228)	X			0.35
Contact Pad to Contact Pad	G	0.45		

Notes:

1. Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

Microchip Technology Drawing C04-23551 Rev A